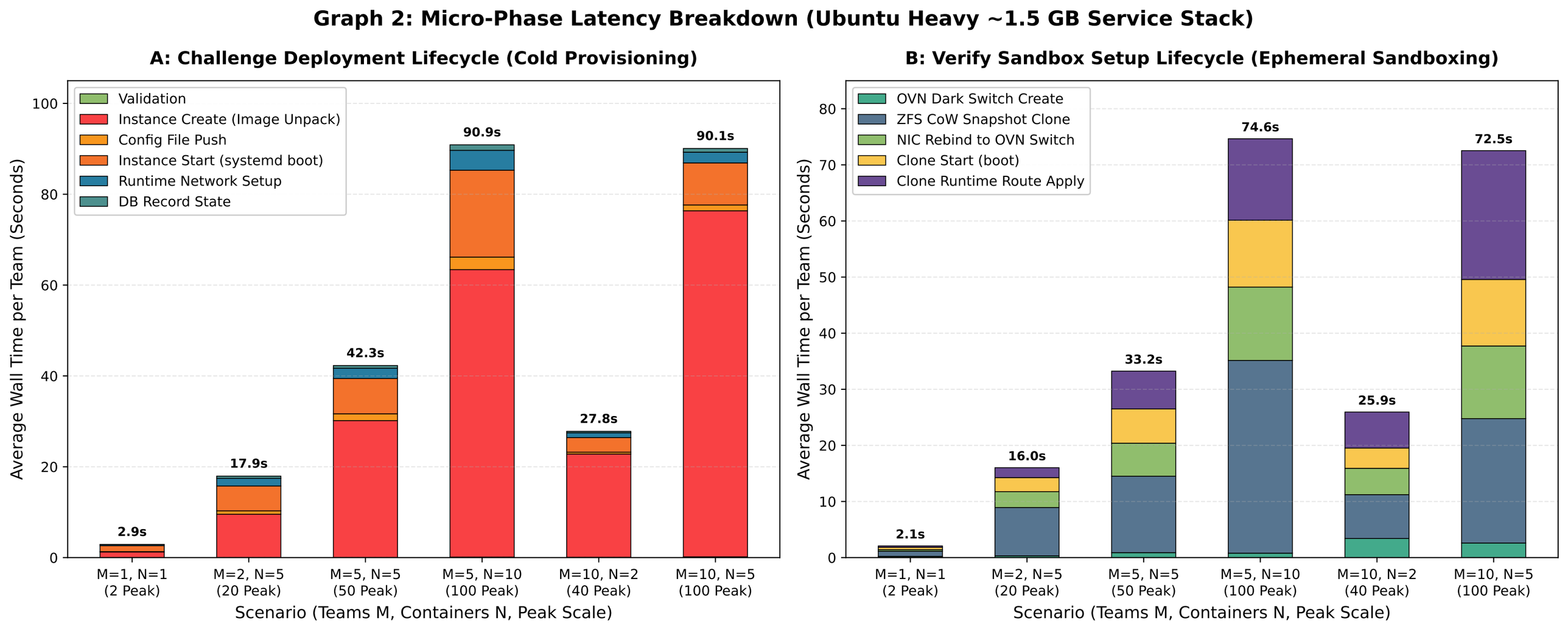
Graph 2: Micro-Phase Latency Breakdown (Ubuntu Heavy ~1.5 GB Service Stack)
A: Challenge Deployment Lifecycle (Cold Provisioning)
Validation
Instance Create (Image Unpack)
Config File Push
Instance Start (systemd boot)
Runtime Network Setup
DB Record State
Average Wall Time per Team (Seconds)
2.9s
17.9s
42.3s
90.9s
27.8s
90.1s
M=1, N=1 (2 Peak)
M=2, N=5 (20 Peak)
M=5, N=5 (50 Peak)
M=5, N=10 (100 Peak)
M=10, N=2 (40 Peak)
M=10, N=5 (100 Peak)
Scenario (Teams M, Containers N, Peak Scale)
B: Verify Sandbox Setup Lifecycle (Ephemeral Sandboxing)
OVN Dark Switch Create
ZFS CoW Snapshot Clone
NIC Rebind to OVN Switch
Clone Start (boot)
Clone Runtime Route Apply
Average Wall Time per Team (Seconds)
2.1s
16.0s
33.2s
74.6s
25.9s
72.5s
M=1, N=1 (2 Peak)
M=2, N=5 (20 Peak)
M=5, N=5 (50 Peak)
M=5, N=10 (100 Peak)
M=10, N=2 (40 Peak)
M=10, N=5 (100 Peak)
Scenario (Teams M, Containers N, Peak Scale)